An Optimal-Shape FPRAS for #Knapsack

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Abstract

The fastest known approximation scheme for counting knapsack solutions is the $\widetilde{O}(n^{1.5}\varepsilon^{-2})$ FPRAS of Feng and Jin (SODA 2025). We show that one change – refining their rounding grid by a factor of n/ℓ, which their own sampler could not afford but a level-enumerating sampler gets for free – drops the total running time to

$$\widetilde{O}\big(n^{1.5} + n\,\varepsilon^{-2}\big)\,,$$

uniformly: the cost of building their data structure, plus the cost of merely writing down the ε^{-2} samples any such estimator reads. Every ε-dependence beyond the sample size is gone.[1]

1 The problem and the result

#Knapsack asks: given item weights $W_1, \dots, W_n$ and a capacity T, how many of the 2^n subsets have total weight at most T? The problem is #P-hard, so one settles for a randomized $(1 \pm \varepsilon)$-approximation (an FPRAS). After two decades of progress (Dyer's $\widetilde{O}(n^{2.5}\varepsilon^{-2})$ [2] through several deterministic and randomized improvements), the record is the $\widetilde{O}(n^{1.5}\varepsilon^{-2})$ algorithm of Feng and Jin [1].

Theorem 1 (candidate)**.** *#Knapsack admits an FPRAS running in time $\widetilde{O}(n^{1.5} + n\,\varepsilon^{-2})$.*

Concretely, at $\varepsilon = 10^{-4}$ and $n = 10^6$: Feng–Jin spend $\sim 10^{17}$ operations; the algorithm here spends $\sim 10^{14}$ – the full factor $\sqrt{n}$ separates them whenever $\varepsilon \le n^{-1/2}$. For $\varepsilon \ge n^{-1/4}$ the bound is flat at $\widetilde{O}(n^{1.5})$, the construction cost alone. The shape is optimal for this framework: one cannot avoid building the structure, and any $(1 \pm \varepsilon)$ ratio estimator reads $\Theta(\varepsilon^{-2})$ samples of $\Theta(n)$-ish description size.

2 Where their time goes

Feng–Jin approximate the count as $|\Omega'| \cdot p$, where Ω' is the solution set of a *rounded* instance – counted exactly by a merge tree of arrays – and $p = |\Omega|/|\Omega'|$ is estimated by sampling from Ω' uniformly and checking true feasibility. Their time splits into

$$\underbrace{\widetilde{O}(n^{1.5})}_{\text{build the tree}} \quad + \quad \underbrace{N \cdot \widetilde{O}(\ell\sqrt{n})}_{N \text{ samples, each a tree query}} \quad , \qquad N = \widetilde{O}\Big(\frac{n}{\ell\,\varepsilon^2}\Big)\,,$$

[1] A Lean 4 development accompanying this memo machine-checks the components introduced here, including the statistical heart (the overcount bound); see the repository.

where $\ell \in [2, 8n]$ is the popular-class parameter of the instance. The two ℓ's cancel: N times the query cost is $n^{1.5}\varepsilon^{-2}$ regardless of ℓ.

Why is N so large? Because their grid step is coupled to T/ℓ^2, coarse enough that the rounded superset overshoots: $|\Omega'|/|\Omega| = \widetilde{O}(n/\ell)$, so p can be as small as ℓ/n, and estimating a rare ratio to relative ε costs $1/(p\varepsilon^2)$ samples. The overshoot is pure rounding debt: every mask's weight is known only up to the accumulated rounding error, and masks inside that error band around T are counted as solutions whether or not they are.

3 The observation

Nothing in the *correctness* of the pipeline forces the coarse grid. What forces it is their sampler: a query reads array segments, so its cost scales with array length, and refining the grid by F multiplies every length – and the whole sampling phase – by F. The coupling is an artifact of *how they sample.*

In earlier work on this repository we replaced their query by a pruned witness-split sampler whose per-draw work enumerates the *witness levels* present at a node – a quantity bounded by the subtree's item count – and never scans array positions. Its per-sample cost is $\widetilde{O}(n)$ *independent of the grid.* That frees the grid entirely:

> *Make the grid fine enough that the rounded count is honest, and pay for it only in array length – which the sampler never reads.*

Refine the grid by $F = n/\ell$, so the total round-down error of any mask stays below one band width T/ℓ. Two things happen at once:

1. Rounding down never loses a solution, so $\Omega \subseteq \Omega'$; and every over-counted mask has true weight within one band width above T. Feng–Jin's own band bound (their Lemma 3.3 – a statement about *true* weights, blind to any rounding) says that band carries only a polylogarithmic fraction of $|\Omega|$. Hence $p = 1/\text{polylog}$ and $N = \widetilde{O}(\varepsilon^{-2})$.
2. Construction grows only linearly in F: a merge costs (length $\times$ $\sqrt{\text{levels}}$), and levels do not change. Summed over the tree the construction is $\widetilde{O}(F\ell\sqrt{n})$, which at $F = n/\ell$ is $\widetilde{O}(n^{1.5})$ – the parameter ℓ cancels here too, now in our favor.

4 The algorithm

1. Run the Feng–Jin reduction and class decomposition unchanged, but set every node's grid step a factor n/ℓ finer than theirs.
2. Build the merge tree with their witness-count subroutine (their Theorem 6.1, which is parametric in array length). Alongside each node, store static per-(level, residual-class) sorted position arrays with prefix counts.
3. Draw $N = \widetilde{O}(\varepsilon^{-2})$ samples with the pruned witness-split sampler: descend the tree splitting each position among the witness levels of the two children; every mass and rank query is a prefix lookup.
4. Check each sampled mask's true weight; output $|\Omega'|$ times the observed feasible fraction (median-amplified).

5 Why this is $n^{1.5} + n\varepsilon^{-2}$

Construction: per merge (length $\times\sqrt{\text{levels}}$) at boosted length $F\ell/2^{h/2}$ and $n_m/2^h$ levels sums, over depths and classes, to $F\ell\sqrt{n} \cdot \text{polylog} = n^{1.5} \cdot \text{polylog}$ at $F = n/\ell$. Sampling: $N = \widetilde{O}(\varepsilon^{-2})$ samples at $\widetilde{O}(n)$ each — level enumeration, grid-independent — is $\widetilde{O}(n\varepsilon^{-2})$. There is no third term.

6 Discussion

What is verified. Beyond the sampler's exactness and cost ledgers, the statistical heart of this result is machine-checked: the sandwich $\Omega \subseteq \Omega' \subseteq \Omega \cup \text{band}$ and the overcount bound $100\,|\Omega'| \leq (101 + 200h)\,|\Omega|$, proved in Lean by composing with our formalization of Feng–Jin's Lemma 3.3.

What is inherited. Their Theorem 6.1 subroutine (used as published, at longer arrays – its statement is parametric in length), the two-phase outer pipeline, and their Lemma 3.3's instance-side hypotheses, discharged by their own reduction.

What remains. At $\varepsilon = \Theta(1)$ the bound is $\widetilde{O}(n^{1.5})$, tying Feng–Jin: the construction itself. Improving it is the open problem they pose in §1.3, tied to the bounded-monotone $(\max, +)$-convolution frontier – and it is now the *only* frontier: sampling, for every ε, is settled at the output-optimal $n\varepsilon^{-2}$. The price paid is space: the boosted arrays occupy $\widetilde{O}(n^{1.5})$ words, the same order as their construction time but more than their unboosted arrays.

References

[1] W. Feng, C. Jin. *A Faster Algorithm for Counting Knapsack Solutions*. SODA 2025. arXiv:2410.22267.

[2] M. Dyer. *Approximate counting by dynamic programming*. STOC 2003.

[3] P. Gawrychowski, L. Markin, O. Weimann. *A Faster FPTAS for #Knapsack*. ICALP 2018.